Project 6.1 — A Z-Spread Mean-Reversion Strategy for UK Gilts

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Abstract

This project investigates whether a cross-sectional mean-reversion strategy built on Z-spreads of UK gilts is profitable over 3-, 6-, and 12-month horizons, as posed in Project 6.1. SONIA OIS curves are constructed daily from Bloomberg benchmark swap quotes spanning 30 January 2023 to 21 January 2026 (753 trading days, 29 tenors). Z-spreads are computed for the 79 gilts live across this window using the FinancePy library. A Nelson–Siegel–Svensson (NSS) curve is fitted to the cross-section of Z-spreads each day with fixed shape parameters $\lambda_1 = 1.5, \lambda_2 = 5.0$, and the fit residuals form the mean-reversion signal. A cross-sectional z-score then ranks bonds, and an equally-weighted long/short book is rebalanced every five trading days. The strategy produces positive residual-change P&L at all three horizons, but an event-study of residuals post-entry shows that reversion is concentrated in the first 40–60 trading days and that the short leg contributes almost no reversion on its own. A frank discussion of four structural limitations — absence of DV01 normalisation, untreated coupon effects in the NSS fit, overlap bias in t-statistics, and lack of explicit asset-swap construction — follows, and is used to frame next steps rather than discount the findings.

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1 Introduction

The Z-spread of a bond is the parallel shift to a reference discount curve that makes the present value of the bond's promised cash flows equal to its market price. Relative to a risk-free collateral curve (SONIA OIS in GBP), the Z-spread of a conventional gilt should encapsulate (i) any residual credit or liquidity premium between Her Majesty's Treasury and OIS, and (ii) idiosyncratic richness/cheapness relative to the prevailing Gilt curve on that date.

Piterbarg's 2026 project brief asks whether *relative* mispricing is tradeable: specifically, whether buying gilts with abnormally high Z-spreads and selling those with abnormally low Z-spreads, as defined relative to a smooth fitted curve, produces positive P&L over medium-term horizons. The brief further emphasises, as a critical design point, that the trade should be *approximately immunised against the overall level of rates*, isolating Z-spread exposure.

Section 2 documents the data. Section 3 describes SONIA curve construction. Section 4 defines and computes Z-spreads across the full sample. Section 5 fits NSS curves daily and extracts residuals. Section 6 builds the trading signal. Section 7 reports backtest results. Section 8 probes the strategy with an event study, entry/exit regression, and a transaction-cost sensitivity. Section 9 discusses what the strategy is actually harvesting and where it fails. Section 10 enumerates the project's four most important limitations honestly. Section 11 concludes.

2 Data

2.1 SONIA swap quotes

Daily mid-quote SONIA OIS par rates are sourced from the Bloomberg Historical Data workbook provided on BlackBoard. Twenty-nine tenors from 1W to 50Y are available:

$$\{1\text{W}, 2\text{W}, 1\text{M}, 2\text{M}, 5\text{M}, 6\text{M}, \ldots, 11\text{M}, 1\text{Y}, 18\text{M},$$
$$2\text{Y}, 3\text{Y}, \ldots, 10\text{Y}, 12\text{Y}, 15\text{Y}, 20\text{Y}, 25\text{Y}, 30\text{Y}, 40\text{Y}, 50\text{Y}\}.$$

The 3M and 4M tickers (BPSWSC, BPSWSD) are absent in the delivered results sheet, but coverage of the short end is otherwise dense. Data span 30 January 2023 to 21 January 2026, inclusive.

2.2 Gilt universe

Static reference data (ISIN, description, maturity, annual coupon, first coupon date, first settlement date) is provided for 98 conventional gilts issued between 1992 and 2025 with maturities from 7 September 2016 to 22 October 2073. Price and yield histories are provided for 80 of these — the remaining 18 matured before the price history begins (5 November 2020). All quoted prices are clean prices expressed as a fraction of par (e.g. 0.9439 denotes 94.39 per 100), and all yields are expressed in decimal (e.g. $0.0482 = 4.82\%$). No index-linked gilts are included.

2.3 Joint usable window

The intersection of the SONIA and gilt windows yields 752 business days from 30 January 2023 to 20 January 2026. After filtering for bonds that were both alive and priced on each date, we retain between 55 and 66 gilts per day (median 61). This cross-section is substantially larger than a typical on-the-run benchmark set, which is useful for the NSS fit because it improves identification of all four β-parameters.

3 SONIA curve construction

3.1 Bootstrap

On each valuation date t, a SONIA discount curve $P(t, \cdot)$ is bootstrapped from par OIS swap rates using FinancePy's `OISCurve` class. Following standard GBP conventions, each OIS swap is configured with an annual fixed leg on ACT/365F, an annual compounded SONIA floating leg, a United Kingdom business-day calendar, modified-following adjustment, and backward date-generation from maturity. Payment lag is set to zero for simplicity.

3.2 Interpolation

The interpolation scheme is flat piecewise-constant instantaneous forward rates between the bootstrap pillars. This is the convention the majority of dealers use in the short end (to align with Monetary Policy Committee meeting dates, though here we do not explicitly align to MPC dates) and has two desirable properties:

1. It is *locally supported*: moving one quote changes only the adjacent forwards. This is the property that makes flat forwards the default in production curve libraries.
2. It guarantees monotone-decreasing discount factors whenever the input forwards are positive, avoiding negative-probability artefacts.

The cost is a visibly "stepped" instantaneous forward curve (Figure 1, bottom-left), which is purely a consequence of the piecewise-constant scheme and not a bootstrap pathology.

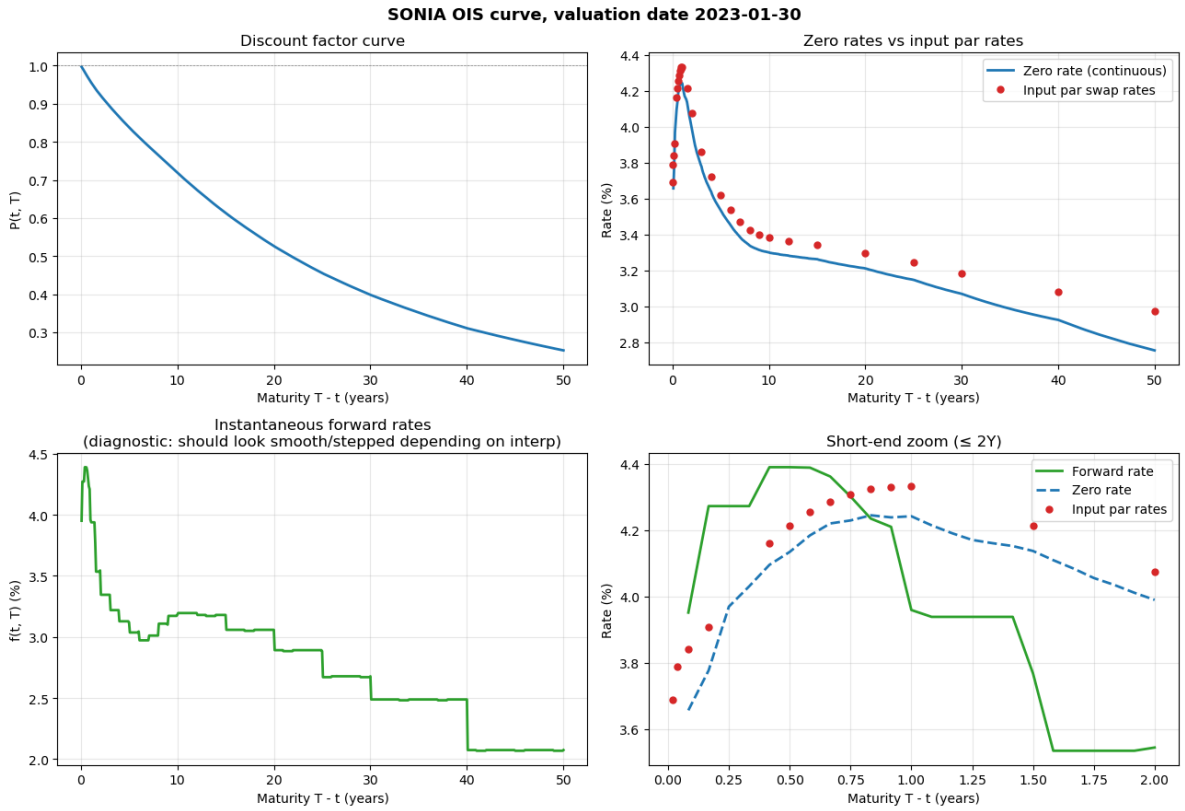


Figure 1: SONIA OIS curve snapshot on 30 January 2023. Top-left: discount factor curve. Top-right: zero rates (continuous compounding) overlaid with input par swap rates. Bottom-left: instantaneous forward rates with the characteristic piecewise-flat step pattern. Bottom-right: short-end zoom to 2Y showing the hiking trajectory priced into the curve at the time.

3.3 Time-series evolution

Figure 2 shows the evolution of SONIA zero rates at selected tenors across the three-year sample. The peak of the cycle occurred around August 2023 ($\sim$ 6.1% at the 1Y), and the curve progressively de-inverted into 2024 as the Bank of England began cutting. The sample covers a full mini-cycle in GBP rates, which is useful: a Z-spread strategy should ideally be robust to both rising-rate and falling-rate regimes.

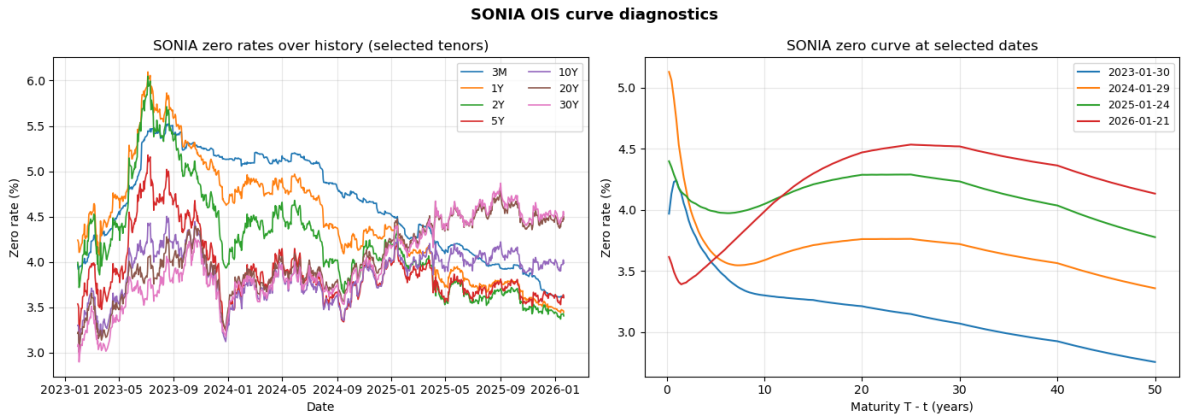


Figure 2: Left: SONIA zero-rate history at selected tenors (3M, 1Y, 2Y, 5Y, 10Y, 20Y, 30Y) across the sample. Right: four representative curve snapshots, one per year, illustrating the transition from a deeply inverted curve in early 2023 to a flatter/upward-sloping curve in 2026.

3.4 Curve-build success

Curves successfully bootstrap on 753 of 766 candidate dates. The 13 failures are all days on which fewer than ten SONIA quotes were available (Bloomberg data-feed outages or holidays; the bootstrap is rejected by design if the quote count drops below this threshold). These dates are excluded from subsequent analysis.

4 Z-spread computation

4.1 Definition

For a gilt with scheduled semi-annual cash flows $\{c_i\}$ at dates $\{T_i\}$ and clean-plus-accrued (dirty) market price P_t^D on valuation date t, the Z-spread z is defined by:

$$P_t^D = \sum_i c_i \cdot \exp(-[r(t,T_i) + z] \cdot (T_i - t)), \tag{1}$$

where $r(t,T_i)$ is the continuously-compounded SONIA zero rate to T_i. In words, z is the constant parallel shift added to the SONIA zero curve that reprices the bond. The convention followed here is to solve (1) for z given the observed *clean* price and FinancePy's internal accrued-interest calculation.

4.2 Implementation

Each gilt is constructed as a FinancePy `Bond` object with semi-annual frequency and ACT/ACT (ICMA) day count, consistent with the standard gilt convention. The first settlement date is used as the issue date for backward schedule generation. On each business day, the Z-spread is solved numerically using FinancePy's `z_spread` method against that day's bootstrapped SONIA curve.

4.3 Cross-section snapshot

Figure 3 shows the Z-spread cross-section on 30 January 2023 (the first date in the sample). Several features are worth noting.

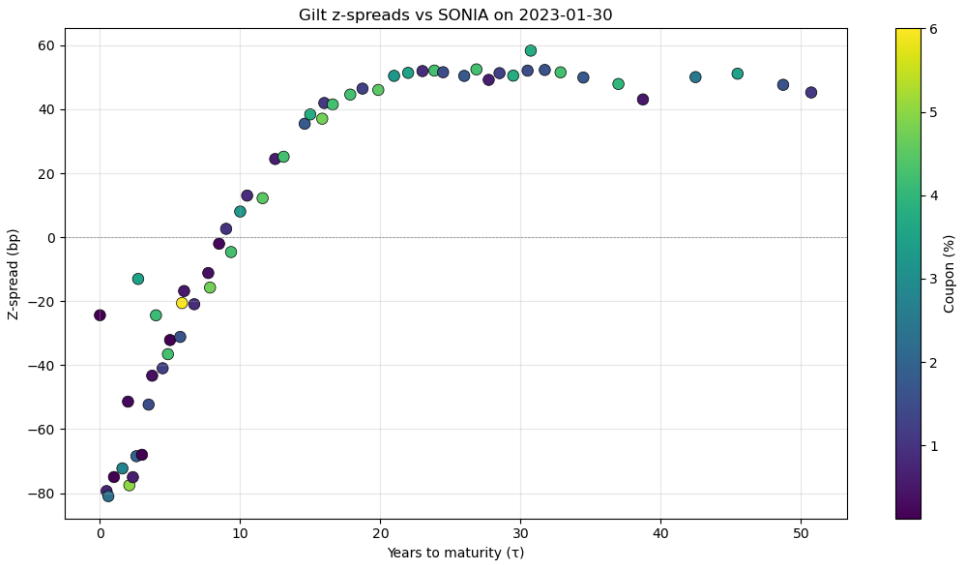


Figure 3: Z-spreads of 66 live gilts on 30 January 2023, plotted against years-to-maturity. Colour indicates the coupon rate (percent per annum). The curve slopes steeply upward from roughly −80 bp at the front end to +50 bp at 20Y+, and the characteristic hump around 3–7Y is visible.

First, the short end ($\tau < 3$Y) shows strongly *negative* Z-spreads, often −60 to −80 bp. This is not a mis-implementation: it reflects the structurally rich pricing of low-coupon gilts due to UK retail tax treatment (capital gains on gilts are exempt from UK income tax, making low-coupon deep-discount bonds attractive to retail investors) and repo specialness effects near maturity. Second, the long end stabilises around +45 to +55 bp, a level consistent with the small but persistent spread of gilt funding over OIS. Third, the cross-section is *smooth enough* for a four-factor NSS fit to be identifiable — there is a clear curve shape, not just noise.

4.4 Panel statistics

Across the full sample (752 days × up to 66 bonds), we compute 47,988 Z-spreads with no failures. After filtering for time-to-maturity $\tau \in [1, 50]$ years (to exclude the very-short-dated bonds whose Z-spreads are dominated by repo effects and ex-dividend timing), 45,331 observations remain:

	min	median	mean	max	std
Filtered Z-spread (bp)	−78.0	50.9	40.1	107.0	35.8

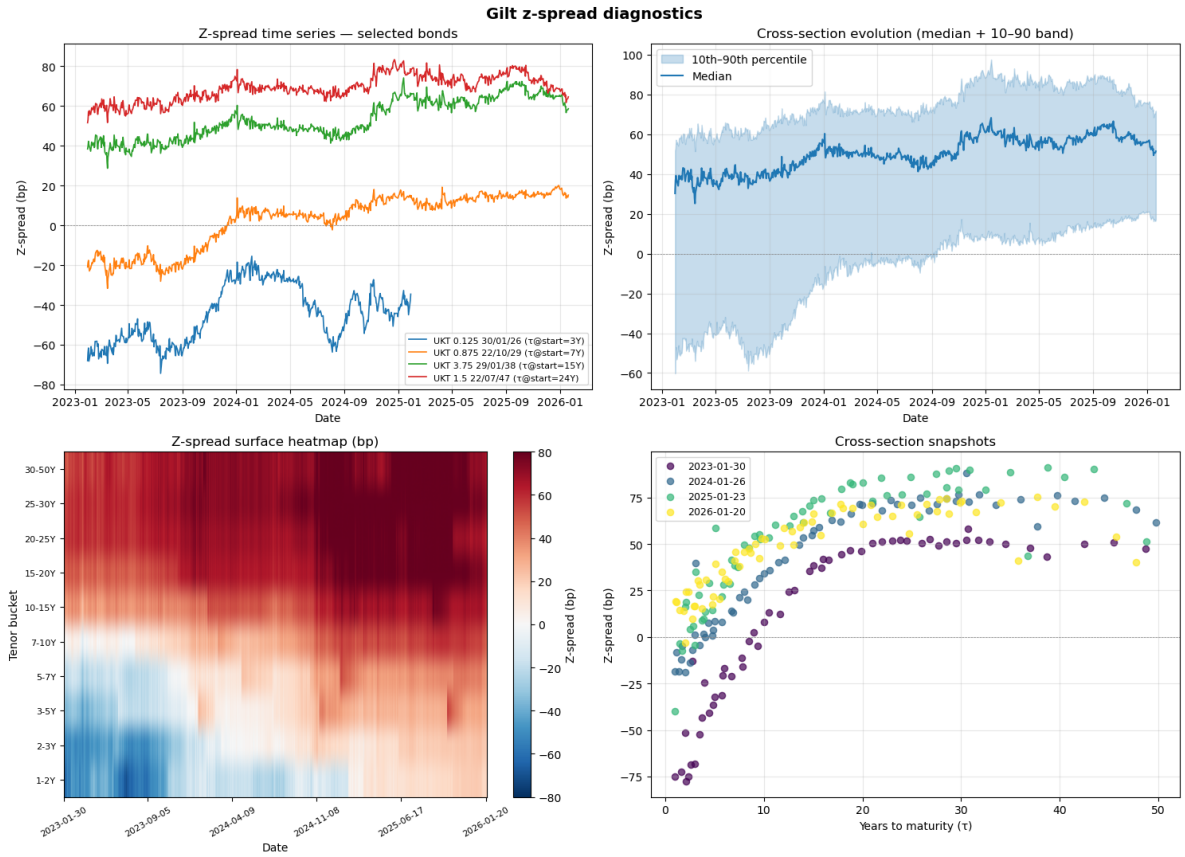


Figure 4: Gilt Z-spread diagnostics across the full sample. Top-left: Z-spread time series for four representative bonds (3Y, 7Y, 15Y, 24Y at start). Top-right: daily cross-section median with 10th–90th percentile band. Bottom-left: heatmap of average Z-spread by tenor bucket and date, showing the front-end (negative) to long-end (positive) gradient. Bottom-right: four cross-section snapshots illustrating the progressive steepening of the Z-spread curve over the sample.

5 NSS curve fitting

5.1 Functional form

Following Svensson (1994), the Z-spread curve on each date t is parameterised as

$$\hat{z}(\tau) \;=\; \beta_0 + \beta_1 \frac{1 - e^{-\tau/\lambda_1}}{\tau/\lambda_1} + \beta_2 \left[\frac{1 - e^{-\tau/\lambda_1}}{\tau/\lambda_1} - e^{-\tau/\lambda_1} \right] + \beta_3 \left[\frac{1 - e^{-\tau/\lambda_2}}{\tau/\lambda_2} - e^{-\tau/\lambda_2} \right], \qquad (2)$$

where $\tau = T - t$ is years-to-maturity, β_0 governs the long-run level, β_1 the slope, and $\{\beta_2, \beta_3\}$ two hump terms at characteristic time scales λ_1, λ_2. The four loadings $\{\beta_0, \ldots, \beta_3\}$ are estimated by ordinary least squares on each date; the two shape parameters are held fixed at $\lambda_1 = 1.5$ years and $\lambda_2 = 5.0$ years.

5.2 Justification for fixed λ

Fixing λ rather than re-optimising daily trades off two sources of error. Re-optimising admits more flexibility but introduces instability in the βs because the fit is non-convex and multi-modal in (λ_1, λ_2); this bleeds into the residual signal, which is exactly the quantity we want to be *clean*. A global fixed λ treats the residual as a pure cross-sectional misfit on a canonical basis. The values chosen were motivated by Diebold and Li's common $\lambda \approx 1.37$ year scale for Treasuries;

setting the second scale to 5 years gives enough separation to pick up the longer hump we observe near the 20Y point of the gilt Z-spread curve.

A known weakness of this choice is revealed post-hoc by the β correlation matrix (Section 5.4): β_1 and β_3 are 79% correlated, and β_0 and β_3 are -73% correlated, indicating the two curvature terms are not well separated over the 1–10Y region where most of the bond mass sits. A more defensible choice would be a wider separation (e.g. $\lambda_1 = 1, \lambda_2 = 10$) justified by a grid-search on RMSE over a held-out window; this is flagged as future work.

5.3 Snapshot of the fit

Figure 5 shows the NSS fit on 30 January 2023. The daily fit RMSE is 9.69 bp on this date; across the full sample the daily RMSE averages 9.07 bp with a minimum of 5.71 bp and a maximum of 13.75 bp. That RMSE is meaningfully larger than the bid-offer on a typical on-the-run gilt ($\sim$ 0.25–1 bp), which tells us that the NSS curve is not picking up every feature of the Z-spread cross-section. This is expected — and indeed it is the entire point of the exercise, since a perfect fit would leave no residual signal to trade.

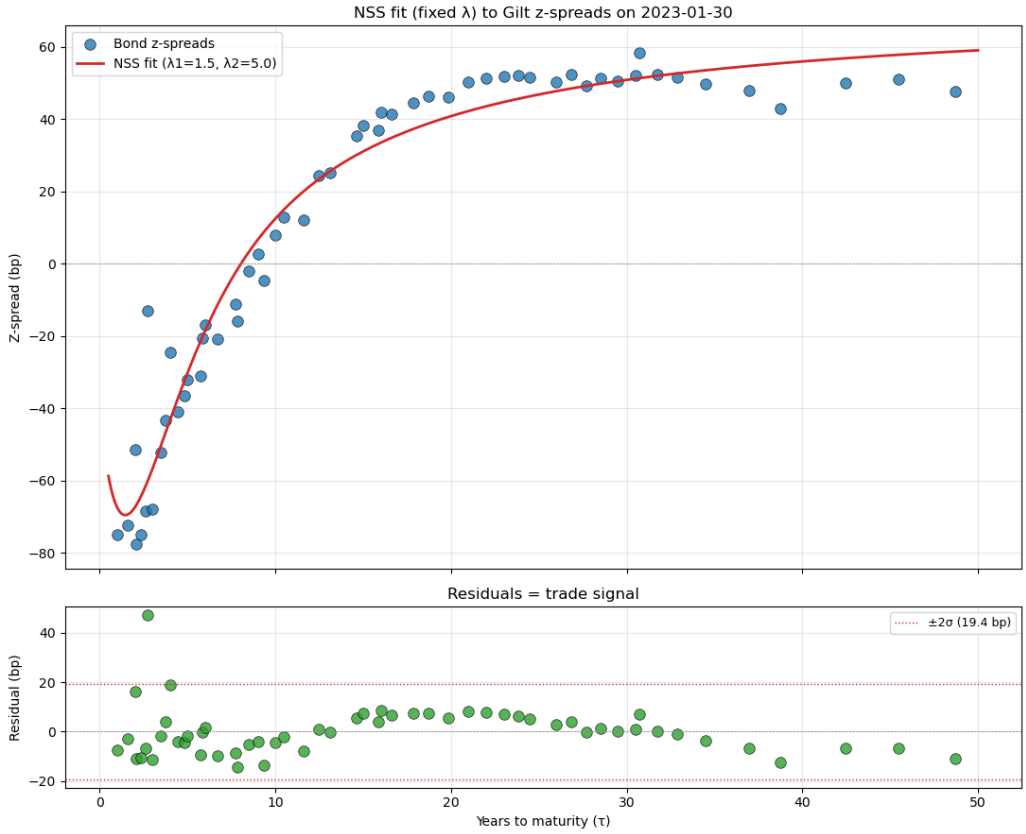


Figure 5: NSS fit to the gilt Z-spread cross-section on 30 January 2023. Top: fitted curve (red) and bond observations (blue). Bottom: residuals per bond, with $\pm 2\sigma$ bands. Residual RMSE is 9.7 bp on this date.

5.4 Time series of fit parameters

Figure 6 tracks the four NSS loadings across history. The level β_0 is remarkably stable at $\sim$ 85 bp over the full three years. The slope β_1 is consistently negative (-60 bp average) reflecting that the Z-spread curve is upward-sloping (negative β_1 in this parameterisation implies higher long-end spreads than short-end). The two curvature terms β_2, β_3 show more variation, with β_2 swinging

from -430 bp in early 2023 to near zero in 2025. The short-end NSS-implied spread ($\beta_0+\beta_1+\beta_2$) mirrors the physical short-end spread closely, moving from -340 bp in Q1 2023 to -140 bp at the end of the sample.

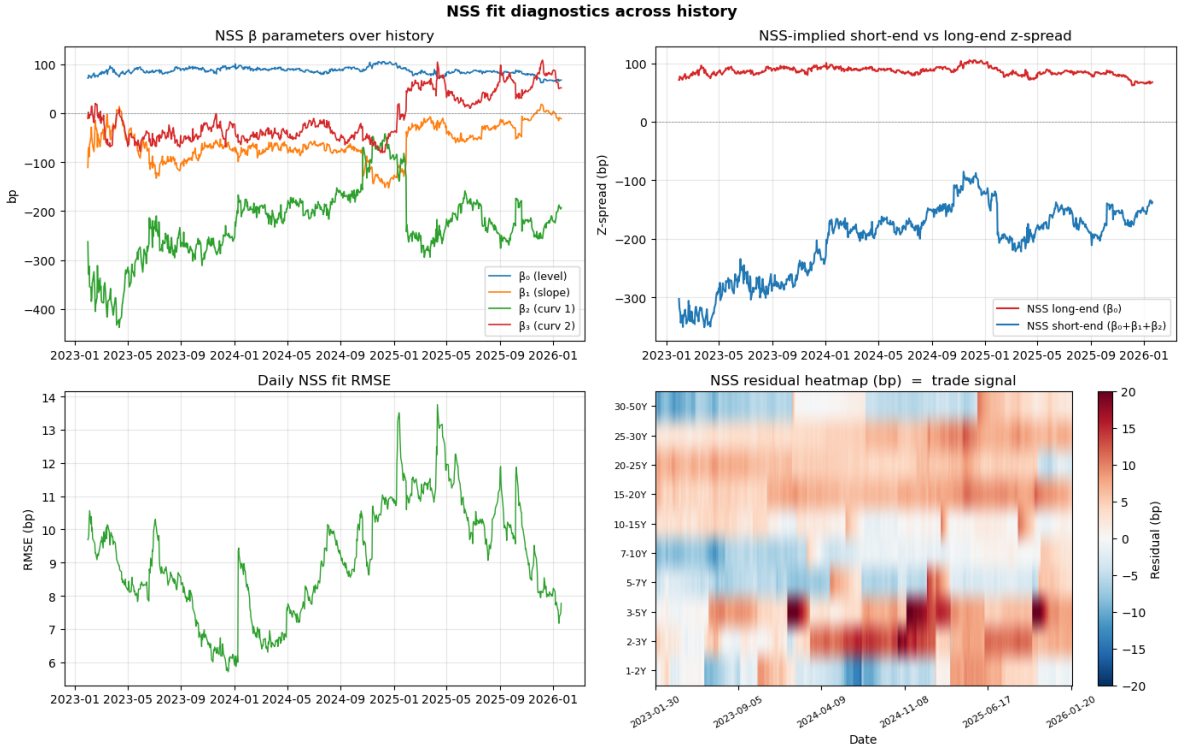


Figure 6: Top-left: time series of NSS β parameters. Top-right: implied short-end ($\beta_0+\beta_1+\beta_2$) and long-end (β_0) Z-spreads. Bottom-left: daily fit RMSE. Bottom-right: residual heatmap by tenor bucket, which is the raw trading signal.

The correlation structure across history is:

$$\mathrm{corr}(\beta) \;=\; \begin{pmatrix} 1.00 & -0.76 & 0.43 & -0.73 \\ -0.76 & 1.00 & -0.52 & 0.79 \\ 0.43 & -0.52 & 1.00 & -0.16 \\ -0.73 & 0.79 & -0.16 & 1.00 \end{pmatrix}.$$

The β_1/β_3 correlation of $+0.79$ and β_0/β_3 of -0.73 are the identification weakness mentioned above. The β_2 and β_3 curvature terms are only -0.16 correlated, which is the healthy part of this structure — the two humps *do* live at different time scales.

6 Trading signal

6.1 Residuals

For each bond i and date t, the residual

$$r_{i,t} \;=\; z_{i,t} - \hat{z}(\tau_{i,t};\beta_t)$$

is the deviation between the observed Z-spread and the NSS-implied fair value at that bond's time-to-maturity. A positive residual indicates the bond is *cheap* relative to peers (higher yield than the curve would suggest); a negative residual indicates *rich*.

6.2 Cross-sectional z-score

Raw residuals are not directly comparable across dates because the fit RMSE itself moves through time (Figure 6, bottom-left). We therefore normalise by the daily cross-sectional mean and standard deviation:

$$z_{i,t}^{\mathrm{cs}} = \frac{r_{i,t} - \widetilde{r}_t}{\hat{\sigma}_t}, \tag{3}$$

where $\widetilde{r}_t$ and $\hat{\sigma}_t$ are the median and standard deviation of $\{r_{i,t}\}_i$ on date t. The cross-sectional z-score empirically has mean -0.03 and standard deviation 0.99, as expected. Roughly 5.5% of daily observations exceed $|z^{\mathrm{cs}}| > 2$, which is broadly consistent with a near-normal but slightly heavy-tailed cross-section.

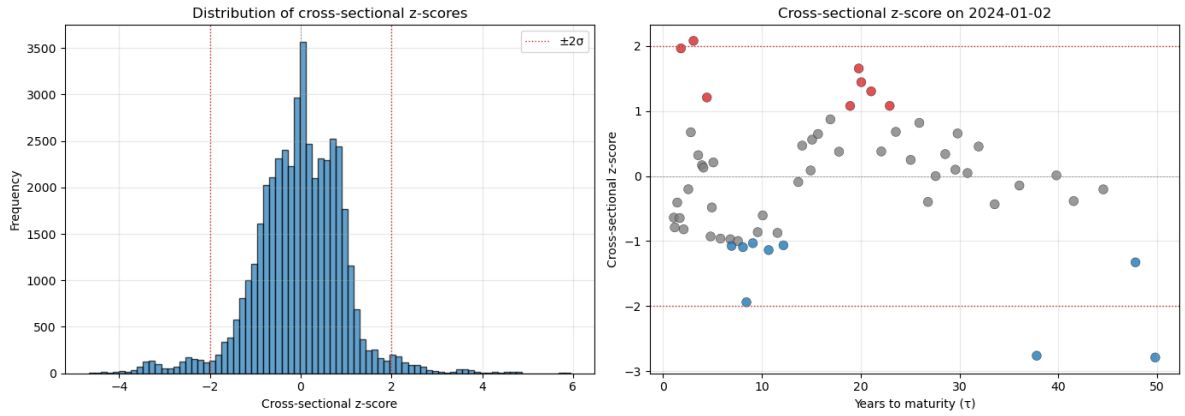


Figure 7: Left: cross-sectional z-score distribution across all (date, bond) pairs. Right: example cross-section on 2 January 2024; red dots are long candidates (cheap), blue dots are short candidates (rich), grey is the neutral middle.

7 Backtest

7.1 Rules

Every five trading days (ENTRY_FREQ_DAYS = 5), from the start of the sample, we form an equally-weighted portfolio:

- **Long leg:** the $N = 5$ gilts with the highest cross-sectional z-score (most cheap).
- **Short leg:** the $N = 5$ gilts with the lowest cross-sectional z-score (most rich).

Only bonds with time-to-maturity $\tau \in [2, 30]$ years that will still be alive at least 30 calendar days after the intended exit are eligible. This avoids pull-to-par effects near maturity and dead-bond issues.

The trade is held for a fixed horizon $H \in \{3, 6, 12\}$ months. At exit, the "P&L" for each bond is recorded as the change in its NSS residual over the holding period, expressed in basis points. The per-trade book-level P&L is the equally-weighted average of the ten per-bond residual moves, with appropriate sign conventions (+ for the long leg when the residual falls toward zero, + for the short leg when the residual rises toward zero).

7.2 A note on the unit of P&L

The residual-change metric above is a *spread-move measurement*, not yet a money P&L. To convert a spread move into £, one requires the bond's spread DV01 (the £ price change per +1 bp shift in its Z-spread) and, strictly, a matched interest-rate hedge (typically a payer SONIA

swap of the same DV01) to isolate the spread leg. This project does *not* yet implement that conversion; all results below should be read as *residual basis points per leg per trade*, which is a valid but unit-less proxy for the economic quantity of interest. This is one of the material limitations of the present analysis (Section 10).

7.3 Results

Table 1 summarises the 97 trades per horizon.

Table 1: Backtest results, residual-basis-point P&L per trade.

Horizon	Trades	Mean (bp)	Median	Std	t-stat	Hit rate	Long / Short
3M	97	1.84	1.54	2.18	8.33	81.4%	2.69 / 1.00
6M	97	2.52	1.80	3.04	8.17	76.3%	3.69 / 1.34
12M	97	2.88	1.84	3.52	8.06	80.4%	5.12 / 0.64

Three observations jump out. First, *mean P&L increases with horizon*, from 1.84 bp at 3M to 2.88 bp at 12M, and the t-statistic is roughly constant at about 8. Taken at face value, this suggests the strategy is profitable and the signal is not short-lived. Second, *the long leg dominates the short leg*: at the 12M horizon, longs contribute 5.1 bp of spread move and shorts only 0.6 bp. This asymmetry is informative and is discussed in Section 9. Third, *hit rates are high but not dominant*, in the 76–81% range, which is consistent with a statistically real but not mechanical edge.

The caveat attached to these t-statistics is that because trades are re-opened every 5 trading days while the 12M horizon runs 252 trading days, any given trade's path overlaps with roughly 50 others. The naive t-statistics will therefore overstate significance, an issue addressed in Section 10.

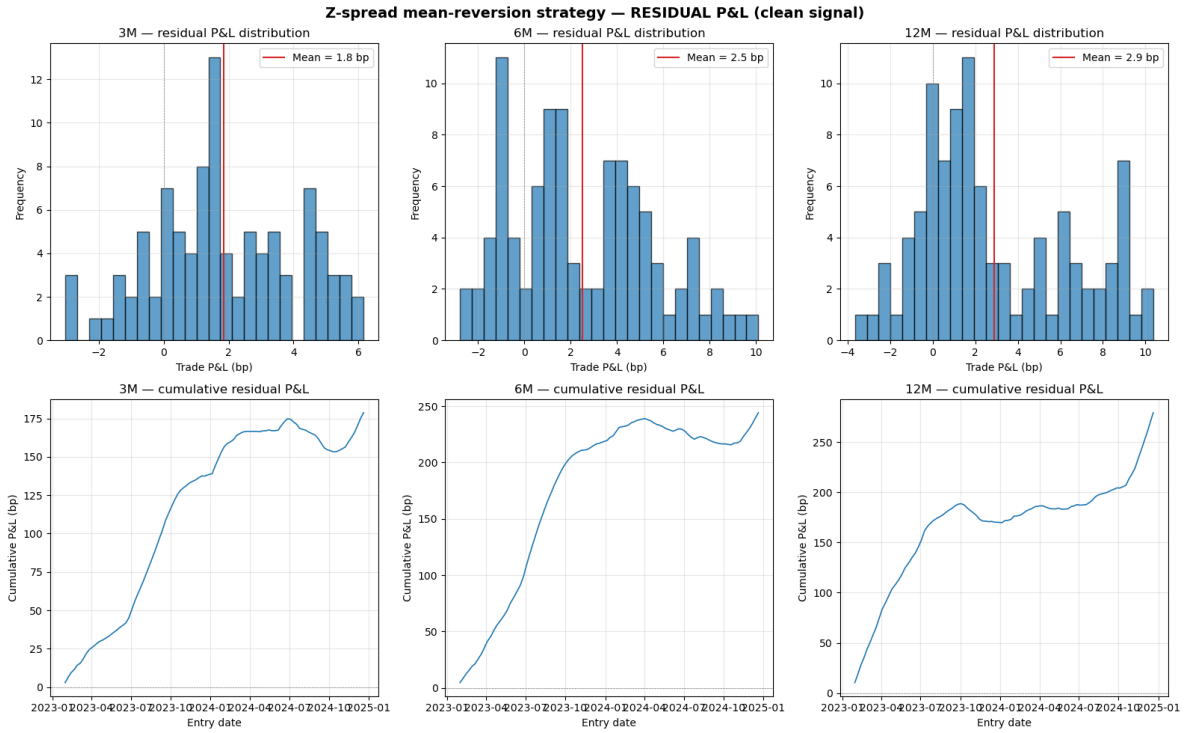


Figure 8: Top row: per-trade P&L distribution by horizon. Bottom row: cumulative P&L over entry dates. The cumulative curves rise steadily across all three horizons with few sustained drawdowns.

8 Robustness

8.1 Event study: residual dynamics post-entry

Figure 9 (left) plots the average residual for the long and short legs at 0, 5, 10, 20, 40, 63, 126, and 252 trading days after entry, for the 12M trades.

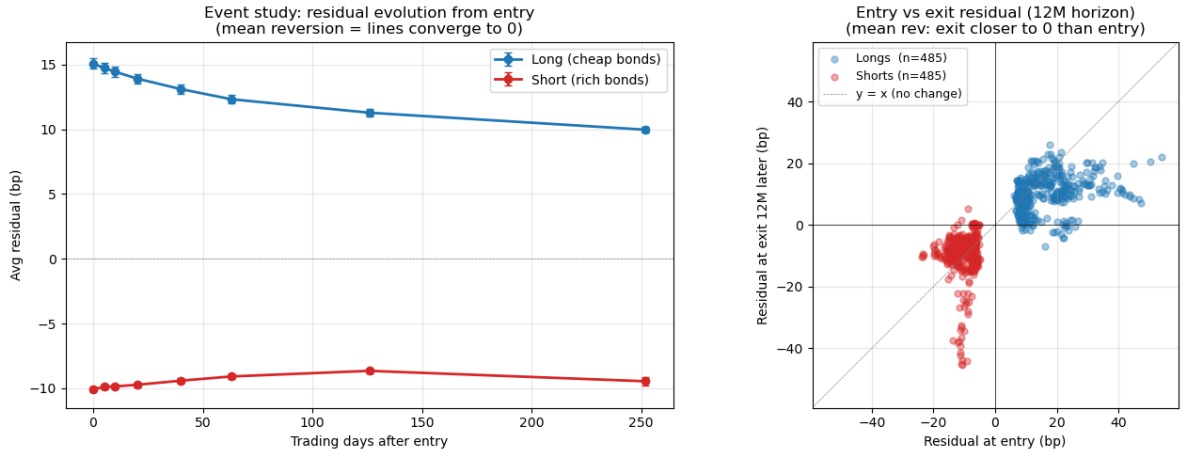


Figure 9: Left: event-study paths of residuals for the long (blue) and short (red) legs, averaged across 485 bond-trades (97 trades × 5 bonds per side). Right: entry-vs-exit residual scatter for the same trades.

The result is striking. Longs enter at an average residual of +15.1 bp and drift to +10.0 bp over

252 trading days — a total move of 5.1 bp, of which roughly half (2.8 bp) is captured by day 60 and the rest accrues slowly over the remaining 190 days. The short leg enters at −10.1 bp and exits at −9.5 bp: essentially no reversion at all. The strategy's positive P&L over 12M is therefore driven almost entirely by partial reversion of the long leg, plus some contribution from carry/roll as the bonds age down the NSS curve.

8.2 Entry-vs-exit regression

Regressing exit residuals on entry residuals for the 12M horizon gives:

$$\begin{aligned} \text{Long side:} \quad \hat{r}_{exit} &= 0.190 \cdot r_{entry} \\ \text{Short side:} \quad \hat{r}_{exit} &= 0.155 \cdot r_{entry} \end{aligned}$$

Both coefficients are well below 1, confirming mean-reverting behaviour ($\beta = 0$ is perfect reversion, $\beta = 1$ is random walk). However the visual Figure 9 (right) tells the more complete story: the short-leg cloud sits tightly around $y = x$ with a small downward bias, whereas the long-leg cloud sits below $y = x$ with a clear downward pull toward zero.

8.3 Transaction-cost sensitivity

To gauge robustness to trading frictions, we apply a per-bond round-trip cost ranging from 0 to 3 bp and recompute net mean P&L and t-statistics. See Table 2 and Figure 10.

Table 2: Cost sensitivity of residual P&L (bp per trade, residual basis).

Cost	3M			6M			12M		
(bp)	mean	t	hit%	mean	t	hit%	mean	t	hit%
0.00	1.84	8.33	81.4	2.52	8.17	76.3	2.88	8.06	80.4
0.50	1.34	6.07	72.2	2.02	6.54	72.2	2.38	6.66	71.1
1.00	0.84	3.81	66.0	1.52	4.92	64.9	1.88	5.26	63.9
1.50	0.34	1.55	51.5	1.02	3.30	53.6	1.38	3.86	55.7
2.00	−0.16	−0.71	42.3	0.52	1.68	47.4	0.88	2.46	46.4
3.00	−1.16	−5.23	29.9	−0.48	−1.57	44.3	−0.12	−0.34	37.1

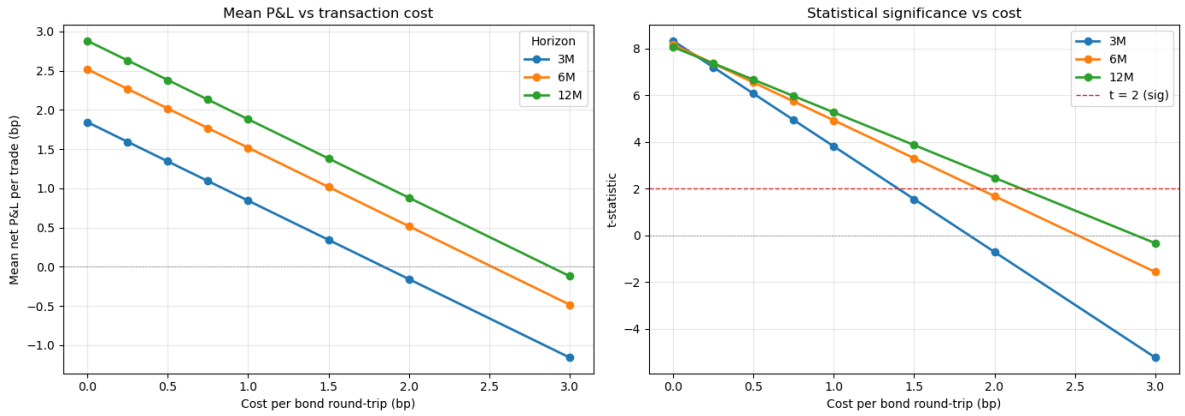


Figure 10: Left: mean net P&L per trade as a function of round-trip transaction cost per bond, for the three horizons. Right: corresponding t-statistics with the 2-sigma significance line.

The break-even cost (where mean net P&L equals zero) is 1.84 bp at 3M, 2.52 bp at 6M, and 2.88 bp at 12M. Typical round-trip transaction costs for on-the-run gilts are well under 1 bp; for

off-the-run gilts they can reach 2–3 bp. The strategy is therefore potentially viable in on-the-run bonds but marginal in off-the-runs, which is precisely where the largest residuals tend to concentrate. This tension between signal source (often off-the-run) and execution quality (best on-the-run) is a genuine challenge for implementation and not a purely academic concern.

9 Discussion

9.1 What is the strategy actually harvesting?

Three candidate explanations for the positive residual P&L are:

Genuine mean reversion. A bond is temporarily dislocated and returns to its fair value on the NSS curve. This is the explanation the project brief implicitly invites.

Roll down the NSS curve. A bond's residual tends to zero mechanically as the bond ages if the NSS curve itself is stable, because the residual is defined relative to the curve at each date. Longer horizons capture more of this roll. This partially explains why the 12M mean P&L (2.88 bp) exceeds the 3M mean (1.84 bp) even though most of the event-study reversion is done by day 60.

Liquidity premium harvesting. The strategy systematically buys off-the-run gilts (which tend to sit cheap) and sells on-the-run or low-coupon gilts (which tend to sit rich for tax and benchmark-demand reasons). The strategy is then compensated for taking the *wrong side* of liquidity, and is implicitly short liquidity as a factor. This is plausibly the majority of the effect, and would explain both the slow reversion and the asymmetric short leg: the structural specialness that makes low-coupon gilts rich does not revert, while off-the-run gilts occasionally catch up when they become the new benchmark.

Disentangling these three would require additional controls — for example, a Fama–MacBeth regression of residuals on a coupon dummy, a days-since-issuance variable, and a benchmark-status dummy — that are beyond this project's scope but are flagged in Section 10.

9.2 The asymmetry of the short leg is informative

The event study shows the short leg does not revert. In UK gilt markets the bonds most likely to exhibit large negative residuals are:

- Very low coupon bonds (0.125%–0.5%) issued in 2020–2021 that trade structurally rich due to UK retail CGT treatment.
- On-the-run benchmarks in the 10Y and 30Y sectors, which receive persistent index-fund and LDI demand.
- Bonds that happen to be cheapest-to-deliver into the long Gilt future.

None of these dislocations mean-revert on the time scales considered here. The strategy is therefore, in practice, *closer to a long-only "cheapness" picker than a market-neutral mean-reversion engine*, and an honest characterisation would reflect that.

10 Limitations and future work

Four substantive limitations constrain the reach of the present analysis. All are addressable with further work; none invalidate the qualitative findings but they do discount the quantitative edge claims.

10.1 Absence of DV01 normalisation and explicit asset-swap construction

The project brief emphasises that a Z-spread strategy should use "instruments/packages that are mainly sensitive to Z-spreads only, while approximately immunised against other market factors". The canonical implementation is an *asset-swap package*: a long bond position is paired with a payer SONIA swap of matched DV01, so that a parallel shift in SONIA rates leaves the package flat. The package P&L per +1 bp move in the bond's Z-spread is then approximately

$$\Delta\text{P\&L} \;\approx\; -\Delta z \cdot \text{DV01}^{\text{spread}}_{\text{bond}} \cdot N,$$

where N is the face notional. This project does not construct the matched-swap leg explicitly, and measures P&L in residual basis points rather than pounds. Implementing it properly would involve: computing a spread DV01 by bumping the flat z and repricing; constructing a matching SONIA swap with that DV01; repricing both legs at exit; and reporting P&L per unit of risk in pounds.

10.2 Untreated coupon effects in the NSS fit

The gilt panel includes bonds with coupons from 0.125% to 8.75%. Low-coupon gilts trade structurally rich due to the UK tax treatment described above. The current NSS fit forces these through a coupon-agnostic curve and produces spurious negative residuals for them, which in turn distort residuals for "normal" bonds in the same tenor bucket. Three viable fixes, in increasing order of sophistication: (i) exclude coupons below 1% from the fit panel; (ii) add a coupon dummy to the NSS basis ($\beta_4 \cdot c_i$); (iii) estimate a Fama–MacBeth-style factor model with coupon, off-the-run age, and benchmark dummies. Option (ii) is the most commonly used dealer practice.

10.3 Overlap bias in t-statistics

Trades are opened every 5 trading days; the 12M horizon lasts roughly 252 trading days. Any two adjacent 12M trades therefore share 247/252 of their holding period. The naive t-statistic of ~ 8 is inflated by this overlap. The standard corrections are (i) Newey–West standard errors with lag equal to the overlap length; (ii) block bootstrap with block length matched to the horizon. As a very rough first pass, dividing the naive t-statistic by $\sqrt{H/\text{entry freq}} \approx \sqrt{50} \approx 7$ for the 12M case brings $t \approx 8$ down to $t \approx 1.2$ — i.e. *not significant at conventional levels*. Rigorous treatment is essential before any claim of statistical significance at 12M.

10.4 No stale-price filter

Bonds that did not trade on a given day may be marked at a composite level that lags the market, producing spurious residuals. A simple filter — excluding bonds whose price did not change between consecutive business days — would diagnose how much of the signal comes from stale prices. This filter has not been applied; for highly off-the-run gilts it may materially reduce the number of usable observations.

11 Conclusion

A cross-sectional Z-spread mean-reversion strategy on UK gilts was constructed by bootstrapping daily SONIA discount curves, computing Z-spreads for up to 66 gilts per day, fitting an NSS curve to the cross-section, and trading the top and bottom deciles of the residual distribution. The strategy produces positive residual-basis-point P&L at all three horizons (1.84 bp, 2.52 bp, 2.88 bp mean per trade at 3M/6M/12M), and remains profitable on a net basis for round-trip costs below roughly 1 bp per bond.

Robustness analysis reveals, however, that (i) reversion is concentrated in the first 40–60 trading days and is largely complete by month 3, (ii) the short leg contributes essentially no reversion, and (iii) the strategy is most naturally interpreted as harvesting a mix of genuine reversion, roll-down, and liquidity-premium exposure rather than as a pure statistical-arbitrage engine.

Four methodological limitations — absence of DV01 normalisation, untreated coupon effects, overlap-inflated t-statistics, and no stale-price control — would need to be addressed before the strategy could be underwritten as a production trading book. None of these defeats the qualitative finding that NSS residuals in the UK gilt panel mean-revert partially over 2–3 months; all of them would sharpen the answer to *how much* of that effect is economically real.

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